Round 2 Report: Integration to GDSII (16–24 Oct)

RoboKnights SemiCon · AXI4-Stream → PCIe 6.0 bridge · Aditya (interconnect, integration), Swaray (link layer), Kshitij (protocol-to-PHY) · Oct 24, 2026 · @Aditya

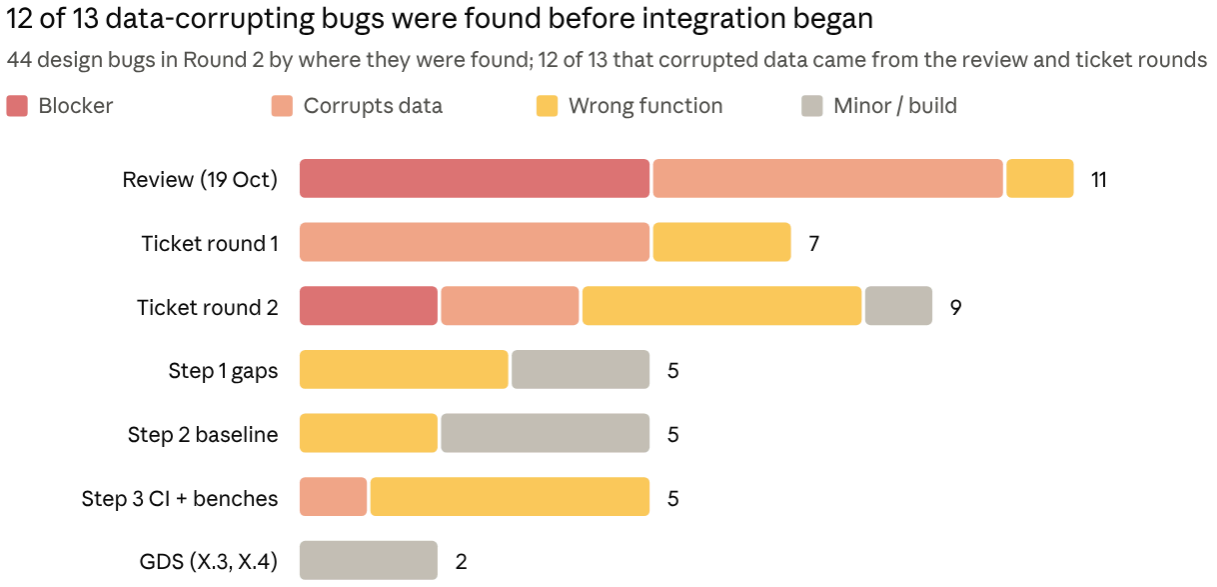
Summary

In eight days the three separately built layers became one verified bridge that meets 255 MHz on FPGA, moves ML tensors at 6.17 GB/s (about 100× the first integrated build) and is going through RTL-to-GDSII on SKY130. On the way, 44 design bugs and 21 testbench bugs were found and fixed; every one is traceable in the project ledger (`claude/SEMICON_LEDGER.md`).

Step	Dates	What it did	Result
Review + tickets	17–20 Oct	Layer-by-layer review of the merged repo; two ticket rounds	27 design bugs fixed, 12 of them data-corrupting; benches repaired
Step 1	20 Oct	Integration gaps G1–G7 (G8 deferred)	Real flow control, host BAR0 writes, XLAT_MISS, shared TUSER header; 144 / 144 TLPs end to end
Step 2	21 Oct	Vivado baseline, perf monitor, first fixes	Baseline 39 MHz → 118 MHz; two stale-copy regressions caught and repaired
Step 3	21–22 Oct	CI, randomized bench, coverage, formal	46 guard rules, 40-seed nightly sweep, 73 / 73 coverage bins, 4 / 4 proofs; found S7, S13, K6
Step 4	22–23 Oct	Three optimisation waves + freeze	118 → 248 MHz; datapath widened to 512 bits end to end
Step 5	23 Oct	In-house traffic analyser	ML 0.06 → 6.17 GB/s; NIC p99 58 → 0.59 µs
GDS	23–24 Oct	LibreLane on SKY130, patches X.1–X.4	Interconnect layout DRC / LVS clean; FPGA 255 MHz with timing met; full-bridge runs in progress

Detailed numbers, charts and the list of deliverables are in Bridge Results and Deliverables; this report tells how the team got there. Swaray's own ledger return is still to come; rows from it will be added where marked.

Where the bugs were found



SEMICON_LEDGER.md v2, BUG rows (excluding TB-BUG), counted by section and severity, 24 Oct

The review of the merged repo and the two ticket rounds did most of the damage control: by 20 Oct all 7 blockers and 12 of the 13 data-corrupting bugs were gone. The one that slipped through, S7 (a payload Flit silently consumed while an FC DLLP was sent), only showed under randomized traffic, which is why Step 3 built the random bench before any optimisation started. Later bugs are narrower: wrong behaviour in one case, two regressions from editing stale file copies, and synthesis-tool rejections in the GDS flow. Testbench bugs (21 more) are not in the chart.

Review and ticket rounds (17–20 Oct)

The three layers were merged into one repository and reviewed layer by layer and at each seam (`AXI4-PCIe6 Bridge Review.html`, 19 Oct). The result was blunt: the layers did not yet form a working bridge.

What the review found (worst first)

Ref	Layer	Problem	Fix
REV-I2	PHY	A second `axi4s_ingest_port` definition: the three layers would not compile together	Kshitij's copy moved to `legacy/`
REV-L1 / L2	Link	Serialiser emitted 15 DWs for a 16-DW packetizer; descriptor FIFO head consumed without pop	Real DW count from TKEEP; gate on serialiser free
REV-L3	Link	Replay buffer had no ACK source: it filled after 31 beats and stopped	ACK / NAK in DLP and control Flits
REV-Seam-C	Link → PHY	No ready; the PHY drained 32× slower than the link sent	PHY Flit FIFO + `phy_tx_ready`
REV-Seam-A	Interconnect → link	TUSER bit order differed: every beat routed as channel 0, address at half the offset	Link-native order, later a shared header (G7)
REV-L4 / L5 / L6 / L7	Link	Sequence number over data bits; TUSER bypassing the replay buffer; 40 bits of every 4th beat dropped; last beat always in slot 3	DLP sequence field; 236 B TLP area in `flit_payload_builder`; contiguous packing
REV-L8	Link	FEC had 8 parity bits and no correction	3-way interleaved SEC-DED + CRC-64
REV-I3	Interconnect / PHY	`session_start` arrived as a level in the wrong clock	`pulse_sync` into `aclk`
REV-I4, REV-L9	Link	1 beat per 20 clocks; one TLP per Flit	Deferred to Step 4 (512-bit engine)

The review also listed missing pieces (composed `pcie_phy_top`, BAR0 register file, MSI-X wiring, 4-lane symbol output) that Kshitij and Swaray built in the same pass.

Ticket round 1 (benches broken by renames). Rewritten or retired 11 link-layer benches, a link `run_all` and file list, and cleaned the PHY's legacy files. Fixing the benches surfaced real bugs: the header parser consumed 2 words for 4-DW headers (SW-H2), header words leaked into payload (SW-H3), the packetizer built DW0 from 31 bits so every header field

sat one bit low (SW-P5), the splitter's registered `s_ready` dropped TLPs (SW-S1), and the write buffer re-read a stale entry (SW-W2). The credit manager was put on the TX path and proven live by `tb_credit_gate_proof` (zero credits stall 197 cycles with no TLP escaping).

Ticket round 2. Nine more design bugs, including a link-layer file that had been overwritten with another module (T2-R1), TUSER order again (T2-R2), the parser address one DW early (T2-R3), FEC `while` loops Yosys rejected (T2-R5) and a 12-bit BAR0 address that truncated the MSI-X table (T2-R7). It also opened the Step 1 gaps (T2-P2-x) and one process ticket, T2-P2-8: "glue edits keep regressing earlier fixes", which became the CI guard in Step 3.

Step 1: integration gaps (20 Oct)

Seven gaps stood between "compiles together" and "behaves like a PCIe device". G1–G6 were made as one patch each, tested together, then replayed as separate commits so each could be applied alone (landed in `5d47645`); G7 is `2380044`.

Gap	What was missing	What was built	Found while doing it
G1	Host writes to the device were not routed to BAR0	Inbound MWr → BAR0 registers (`inb_mwr_*`); `tb_g1_host_bar0`	—
G2	Credit manager fed a fake "infinite" value; no UpdateFC	Real flow control: InitFC1 / InitFC2 / UpdateFC DLLPs and the FC init state machine	Exposed G5a: a bench's "zero credits" was really an InitFC, where 0 means infinite
G3	Address-translation misses vanished	Sticky XLAT_MISS counter at BAR0 0x1C; requests during a stall now queued instead of dropped	G3a: the queued request was dropped on the cycle the stall cleared
G4	MRd decoded as MWr	`parsed_has_data`	—
G5	`tb_tx_engine` printed but never checked	Order monitor, checker task, RESULT line; mutation-tested by breaking Posted priority	G5a–c: three bench bugs, including one counter used both to record and to check

Gap	What was missing	What was built	Found while doing it
G6	One loop integer shared by three loops (Yosys conflicting drivers)	Each loop in its own named block	G6a: SystemVerilog-style inline loop variables pass Icarus but fail Yosys
G7	TUSER layout typed by hand in 8+ files	`shared/axis_tuser_defs.vh` with `tuser_pack()` and a `$fatal` width check; benches use it too	Patch paths regenerated with `git diff` after `git apply` failed
G8	RX decode 1 DW per clock	Investigated only; done in Step 4 as S4-8	—

After Step 1: link suite 16 / 16, `tb_bridge_top` 144 / 144 TLPs and 6 / 6 MSI-X, `tb_bridge_ovf` 39 / 39, and Yosys elaboration of the full bridge (56 files) with 0 latches, 0 conflicting drivers, 0 implicit nets. The G6a lesson (named-block loop scopes, never SV inline declarations) came back three days later in `link_layer_rx` as X.3.

Step 2: baseline and first fixes (21 Oct)

Baseline. Vivado scripts that print a one-page summary per run (part xcu250, Vivado 2026.1), a `perf_monitor` in the bridge bench and the include-path fix the G7 header needed in the Vivado, Quartus and LibreLane flows. The first full place-and-route came in at **WNS −21.45 ns (about 39 MHz)**: every worst path was the CRC / FEC computed in one cycle over a whole Flit in `link_layer_tx`.

First optimisations (ranks 4 and 5).

- **S2-OPT-4:** CRC / FEC computed one 256-bit chunk per clock (9 cycles instead of 1), bit-exact against the reference functions on 8 vectors. Back-to-back Flits went from 10 to 18 clocks, a cost Step 4 later removed.
- **S2-OPT-5:** memory arrays moved out of reset blocks so Vivado infers RAM: replay buffer (60,416 bits), write buffer, RX FIFO, reorder buffer, and the interconnect FIFO (which stays in LUT RAM because its read is first-word-fall-through). Before: 0 BRAM and about half of 116,932 flip-flops were buffers.
- Items 6 (error counters in BAR0) and 8 (wider RX decode) were deliberately deferred; both were done in Step 4 (S4-6, S4-8).

Bugs fixed. A teammate's SymbiYosys proof of the TLP header parser found two decode bugs: MRd seen as MWr (already G4) and locked completions (CplLk / CplDLk) not treated as completions, which raised `malformed_tlp_err` (F6-FIX). The ingest overflow

check compared in 24 bits and could wrap within 64 B of 2^24; F1-FIX widened it and added a zero-length TLAST marker so the link layer closes an aborted session, a gap open since Stage 1.

Two regressions from stale copies. Two fixes were applied to old copies of files kept in the project rather than to the repository. One (S2-REG-1) dropped the G1 / G3 ports from `pcie_phy_top.v`; the other (S2-REG-2) re-derived the F6 change on the 19 Oct baseline and removed `parsed_hdr_done` / `REQ_ON_HDR_DONE` along with G1, G3 and parser fixes. Nine earlier fixes were lost in total. They were restored in `rank45_repair.patch`, and the two root causes were fixed for good: Step 3′s guard checks that every past fix is still present, and on 22 Oct 78 superseded project copies were deleted after checking each fix was in the repository (82 → 4 docs).

Step 3: verification infrastructure (21–22 Oct)

Step 3 built the safety net before any optimisation was allowed, so every later change could be checked automatically.

Piece	What it does
Guard (`ci/check_protected.py`)	A list of lines each earlier fix must keep in the code; fails the build if one disappears. 33 rules at first, 39 with K6, 46 now. It caught all 9 fixes lost in the Step 2 regressions.
`regression.yml`	Every push and PR: guards plus one job per suite (link 17, PHY 25, cocotb 10, interconnect 4, bridge 3, error injection, ASIC buffer sizes, randomized seeds). Nightly: the 600-TLP long run, a 40-seed random sweep, merged coverage, and formal.
`main-guard.yml` + pre-push hook	A broken `main` is reset to the last good commit automatically, with an issue naming the commit and author; direct pushes to `main` are blocked locally.
`tb_bridge_rand` (F1–F9, F7b)	Random sessions, beat counts, last-beat sizes, gaps, weights, unaligned bases, PAM4 noise (F1–F4); overflow at a random point, lane reversal, hot reset mid-session (F5–F7); MSI-X masking and host BAR0 writes (F8–F9); cable pull and glitch (F7b).

Piece	What it does
Coverage	One `COV:` line per run, merged into a table of 73 required bins. On the final code: **73 / 73 over 40 seeds**, including 1,069 FEC corrections, 262 replays, 10 hot resets and 7 cable pulls with correct data.
Formal	SymbiYosys: Gray inverse, PAM4 map against its spec, start-of-Flit strip, TLP header parser. **4 / 4** on the final code.

Bugs the new checks found

Ref	Found by	Bug	Fix
S7	Replays on a noise-free random run	`link_layer_tx` consumed a payload Flit while sending an FC DLLP Flit: payload lost, recovered only by replay	Hold the payload Flit (Swaray). Independent re-run: replays 1 → 0, `tb_bridge_top` 8.3 µs faster
S13	F9 (host BAR0 writes)	A host write to the device's own BAR0 was counted as a translation miss	Exclude the BAR0 window from XLAT_MISS
K6	Ticket from the integration review, needed before F7b	Once in L0, only a reset left it: on signal loss the link stayed "up" and data stopped silently	Recovery-lite in the LTSSM: exits on signal loss (64 clocks), lost framing (512 unframed cycles) or partner TS1s; retrains in ~1,700 clocks (Kshitij)
KS-K6-IDLE	Kshitij, while doing K6	TX lanes went quiet whenever there was no Flit, so "no signal" could not mean link loss	Logical idle in every free L0 slot, as in real PCIe
KS-K6-RST	Kshitij, while doing K6	After a link drop the deskew lock, lane-reversal latch and link-layer state only cleared on reset, so the link could not retrain	3-clock datapath reset pulse on link-down

Ref	Found by	Bug	Fix
S12	Kshitij's review of the F6 fix	Locked completions had no test; a project copy's test passed for the wrong reason	Tests 7a / 7b; reverting F6 makes 7a fail (Swaray)

F7b (cable pull and glitch in the random bench) was written after K6, to test that recovery end to end.

Step 4: optimisation (22–23 Oct)

Target: 250 MHz on the PCIe clock (4 ns), xcu250, Vivado 2026.1. Work was ordered by the failing-endpoint histogram from each run, in three waves; one requirement per commit, full regression before each.

Wave	Changes	Fmax after
Start (`step4-base`)	—	118 MHz (19,046 failing endpoints)
A	RX CRC / FEC check in 3 stages (S4-T1); TX framer 3-stage pipeline, 8 clocks per Flit instead of 18 (S4-T3); sticky error counters in BAR0 0x40–0x64 (S4-6, Kshitij's layer); per-channel session limit, `session_end` , interleaved weighted round-robin (S4-7, Aditya's layer)	—
B	512-bit TLP engine `tlp_tx_wide` replacing serialiser + 32-bit packetizer + packer (S4-1); merging beats into max-payload TLPs, never across 4 KB (S4-2); wide PHY, 8 B per lane per clock (S4-3, Kshitij's layer); 16-DW per clock RX delimiter and header decode (S4-8)	**49 MHz**: widening put the unrolled CRC-64 in one 51-level cycle
C	CRC-64 split into per-chunk parts; merge unit sized from registers; replay timer 2,048 clocks for the wide PHY	147 MHz

Wave	Changes	Fmax after
Freeze (`step4-final`)	CRC / FEC syndromes as explicit XOR trees from constant masks; inbound beat registered before header gather	248 MHz (82 endpoints, −0.027 ns)
+ X.1, X.3, X.4	Ingest TREADY no longer depends on TKEEP; FEC correction as constant-position gates	**255 MHz, all endpoints met** (commit `e29db76`)

Area moved from 56k LUTs / 24k flip-flops to 57k / 39k: the logic grew little despite the datapath going 16× wider, because the 32-bit serial path it replaced was removed; the flip-flops are the new pipeline stages. No BRAM, URAM or DSP is used. The chart and per-run worst paths are in the results doc.

Step 5: load analyser (23 Oct)

Timing alone does not show whether the bridge serves its two users, so the team built an in-house traffic analyser on the full bridge. `tb_bridge_rand` gained workload modes (`+WL=1..5`): SmartNIC IMIX (64 / 596 / 1520 B at 7:4:1), back-to-back 64 B packets, ML tensors from 64 KB to 16 MB, tensors starting just below a 4 KB line, NIC + ML mixed at weights 1:1 / 1:3 / 1:7, and channel noise at 300 and 1000 ppm. Each run prints one `LOAD:` line (throughput, packet rate, p50 / p99 latency, buffer peaks, replays, FEC fixes, credit stalls); `ci/load/run_workloads.py` runs the scenarios and writes JSON, `compare.py` gates each commit against the committed `step4-base` results (2% throughput, 5% p99, 10% replays), and `load.yml` runs it in CI.

Workload	Base	Final	Change
ML tensors (64 KB)	0.06 GB/s	6.17 GB/s	~100×
SmartNIC IMIX	0.15 Mpps, p99 58 µs	4.84 Mpps, p99 0.59 µs	32× rate, 99× lower tail
Back-to-back 64 B	0.48 Mpps	11.6 Mpps	24×
Mixed, NIC:ML 1:7	NIC p99 241 µs	NIC p99 4.2 µs	57× lower
Mixed + 300 ppm noise	~0 GB/s	~0.1 GB/s	still replay-bound

The gains come from Wave B (512-bit engine, merging, wide RX and PHY); the timing waves after it left behaviour unchanged, which a rerun on the final code on 24 Oct confirmed scenario for scenario. The analyser also showed that the arbiter weights do what they say (raising ML to 1:7 moves bandwidth to ML at a measured cost to the NIC tail) and that heavy channel noise remains the weak spot: replays, not bandwidth, limit throughput there. The fairness metric (Jain) was taken out of the gate because a weighted arbiter is unfair by design.

GDSII phase (23–24 Oct)

Flow: LibreLane (OpenROAD-based) on the SKY130A PDK with `sky130_fd_sc_hd` cells, two asynchronous clocks. Each fix is a numbered patch (X.n) applied separately to the FPGA line and the GDS branch.

Patch	Why	What
X.1	The interconnect run failed setup by −12 ns: at the slow SKY130 corner, TKEEP → popcount → offset add → limit compare → TREADY → FIFO write took ~27 ns in one cycle	Offset kept as `off_base + bytes_q`; full beats accepted without looking at TKEEP; a beat within 64 B of the limit is held one cycle while its size is checked. Bench and guard rules updated
X.2	The full bridge (wide PHY, 32-Flit replay buffer) is too large to place and route in the time left	Buffer-size parameters plumbed through `bridge_top` (defaults = FPGA build); a narrow GDS build (narrow PHY, replay 4, FIFOs 4 / 8, 30 ns) with a matching simulation profile in CI
X.3	LibreLane's Yosys reported 230 errors in `link_layer_rx`: latches and logic loops in the FEC correction block, and one loop integer driven from two blocks (the G6 pattern again)	Defaults at the top of the block; a separate index for the clocked loop
X.4	A newer Yosys still found 192 logic loops in the loop form	Each protected bit's Hamming position made an elaboration-time constant; correction as plain gates. Same results on the error-injection bench (19 FEC fixes, 4 replays); a 2-minute `rxcheck` config proves the synthesis checks pass

Flow fixes: linter off (Verilator rejected the cell models' timescales), explicit file lists (testbenches in `src/` broke synthesis), exact port names in the SDC, and running from the repository root when it lives outside `$HOME` (Docker mounts only home and the current folder).

Results so far

- **Interconnect layout (before X.1):** GDS produced; DRC, LVS and antenna clean; hold met; 1.20 mm² die at 41% utilisation, 49k cells (9.8k flip-flops), 104 mW. Setup short by 0.95 ns at the slow corner (0.52 ns after X.1).
- **`link_layer_rx` alone after X.4:** synthesis checks clear; 1.12 mm² of cells.
- **Full bridge:** three runs in progress (narrow build at 30 ns; same at 50 ns with timing repair off as a safety net; the full FPGA design at 50 ns as a stretch). Results will be added when they finish.
- **FPGA with the X patches:** 255 MHz with every endpoint met (the freeze plus X.1, X.3, X.4 went from 248 to 255 MHz).

Open items and lessons

Open

Item	Detail	Owner
Full-bridge GDS	Three runs in progress; results to be added	All
Heavy channel noise (KS-NOISE-1)	At 2,000–3,000 ppm, far above CI noise, the MSI-X check (F8) or the host BAR0 write check (F9) can time out; also seen on the Step 3 baseline. Data is never corrupted (every Flit is CRC-64 checked). Throughput at 300 ppm is ~0.1 GB/s	Swaray / Aditya
Guard rule comments (KS-GUARD-CMT)	A guard rule whose text contains `//` can never match	Aditya
PHY checker (KS-CHK-P3)	A reset-controller checker prints `VIOLATION` lines the runner does not count	Kshitij
Swaray's ledger return	His own list of Round 1 and Round 2 fixes, to be merged	Swaray

Lessons

1. **Edit the repository, never a copy.** The worst setback of the round (nine fixes lost in Step 2) came from applying changes to stale copies. A guard that checks every past fix is still present, plus deleting the copies, stopped it from recurring.
2. **Review before integrating.** The 19 Oct review and ticket rounds removed all blockers and 12 of 13 data-corrupting bugs before any integration work, which kept Steps 1–5 to narrower bugs.
3. **Random traffic finds what directed tests miss.** S7, S13 and the need for K6 surfaced only once the random bench, coverage and cable-pull tests existed.
4. **Write for every tool from the start.** Code Icarus and Vivado accepted (inline loop variables, procedural correction loops) was rejected by Yosys in the GDS flow twice (G6a, X.3 / X.4). Running a synthesis check early would have caught it.
5. **Measure the user's traffic, not just timing.** The load analyser showed the 100× throughput gain and the remaining noise weakness that Fmax alone would not.